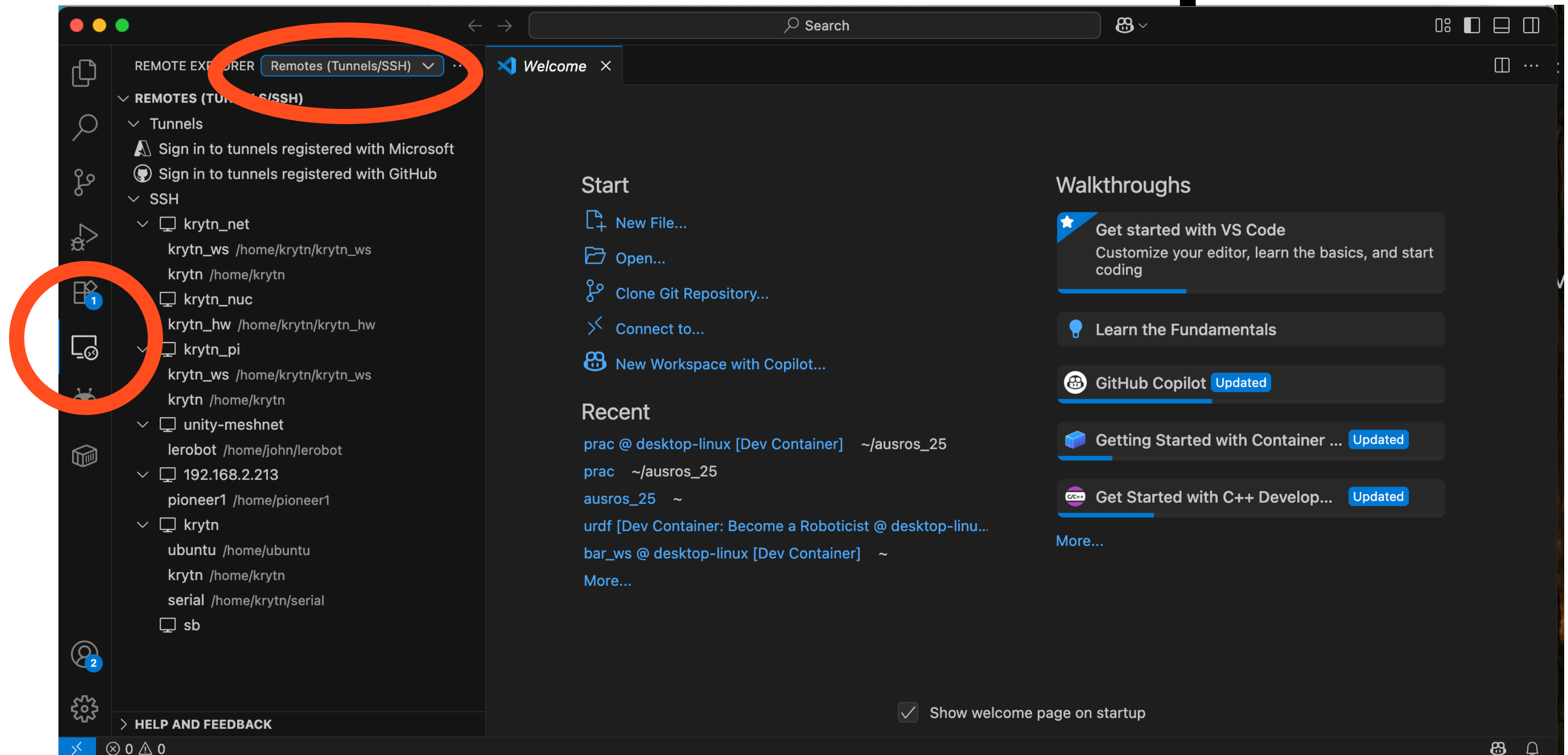


Using LLM's

How to build ROS apps with LLM's

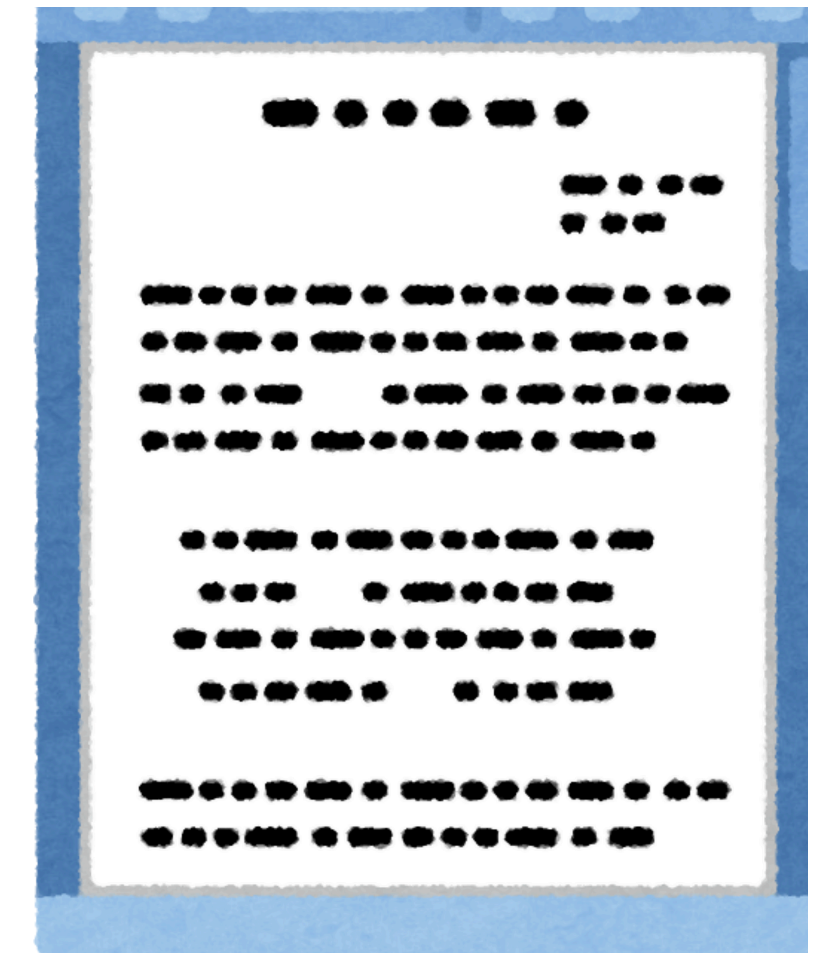
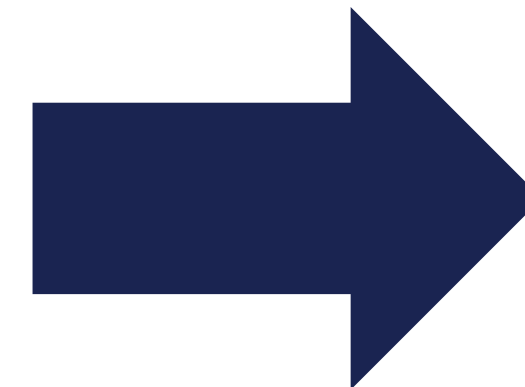
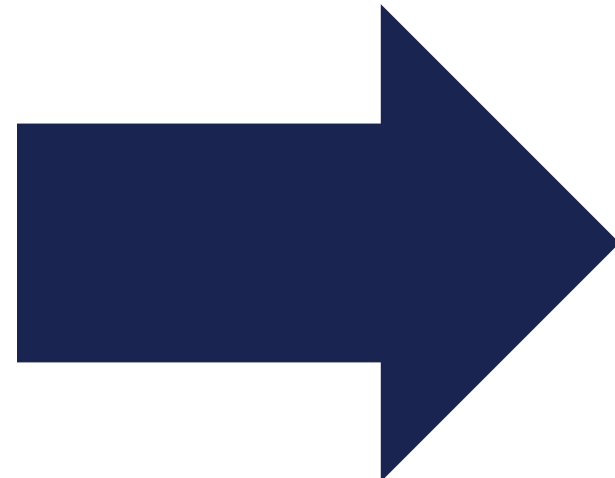
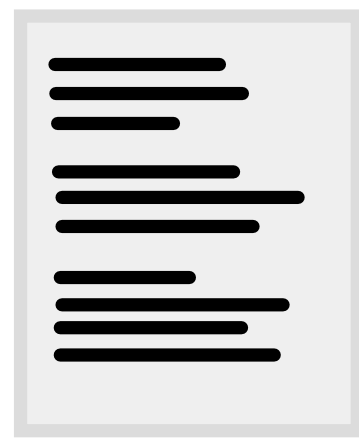
VS Code & Remote Explorer



Large Language Models

VS Code is just a convenient interface to this!

Completion of the request.



Your:

- Code
- Designs
- Errors
- Style
- Requests
- Architecture

Some Linear Algebra

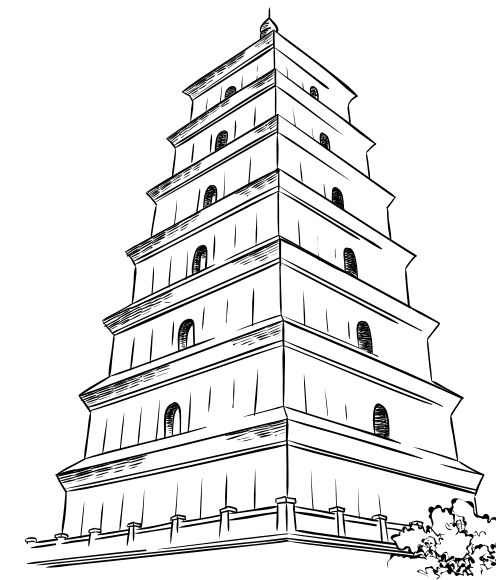
Context Engineering 101



AI is smart but needs you to be very specific about what you need.



It's always day 1 for the AI.
It won't remember anything.



VS



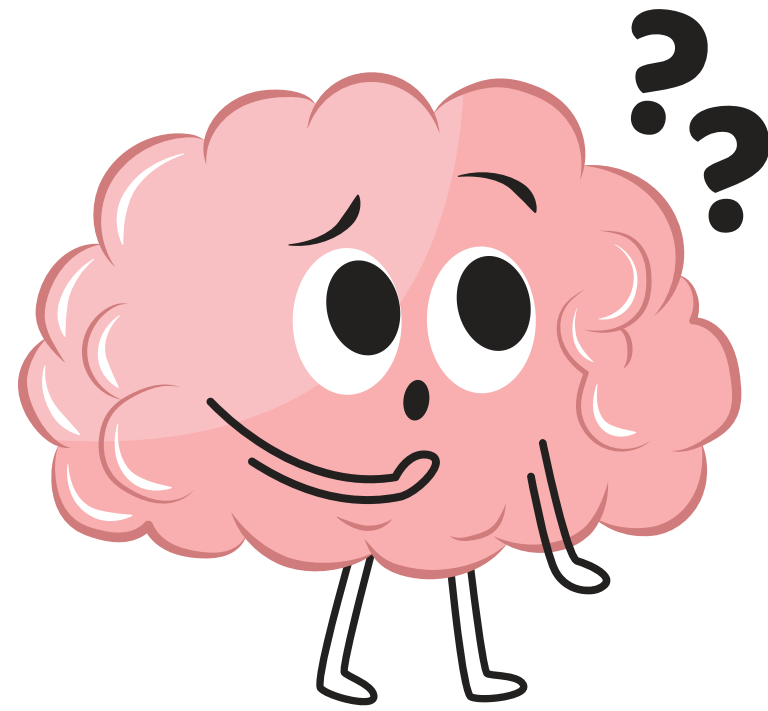
You need to tell it what code style and standards you want.

3 Common Mistakes



AI will always want to agree with you.

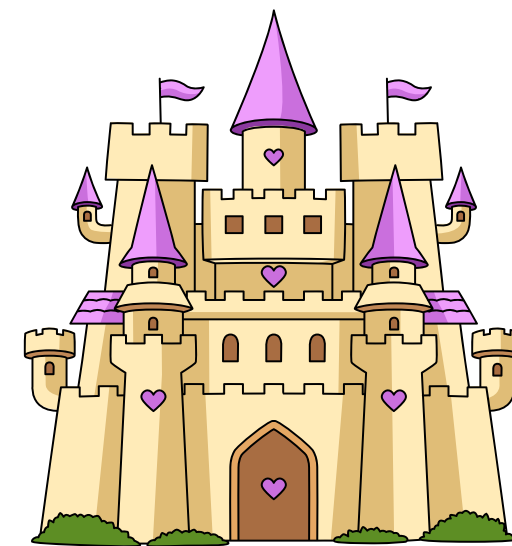
“There is a bug in this code please fix”



AI is easily confused.

Keep your chats or bug fixes to a single request. Use context files to guide.

“Make a castle”



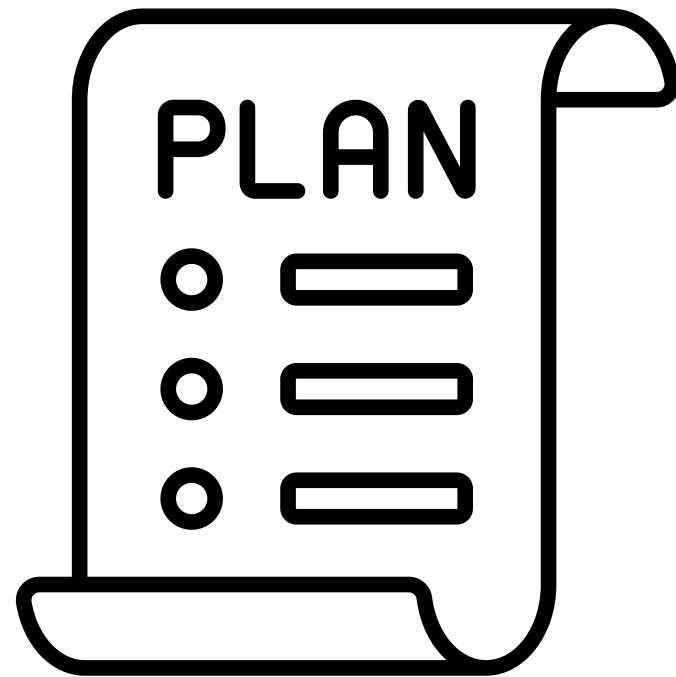
OR



Avoid “vibe” coding.

You need to stay in the loop, otherwise you might get something unexpected and broken.

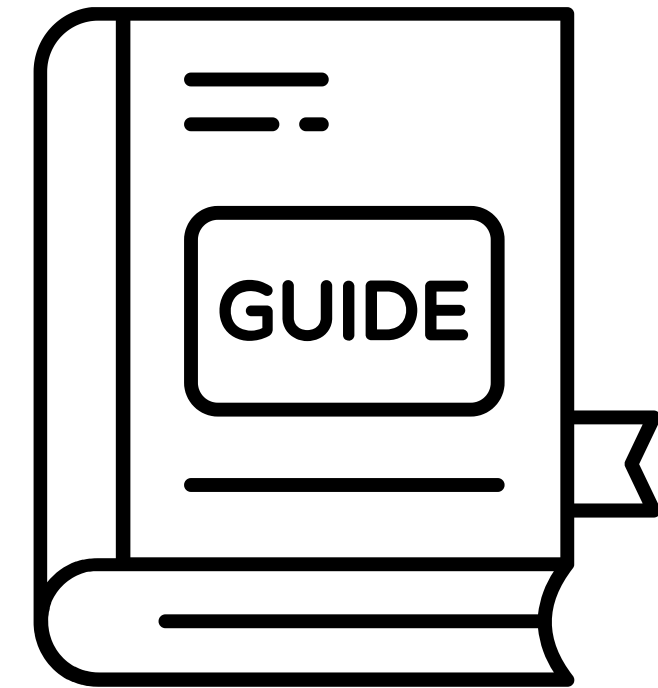
2 Key Context Files



overview.md

Write down your project plan. Break it into small steps and milestones.

Tell the LLM what part of the plan you are currently working on.



architecture.md

Explain to the model what custom preferences you like, or are using.

I.e. Using ROS 2 humble, writing in Python, prefer library calls over.
Include code examples

Ask



Use to understand something

You are in complete control.

Edit



Allow AI to edit just one file

You will approve the edits, get to see the code.

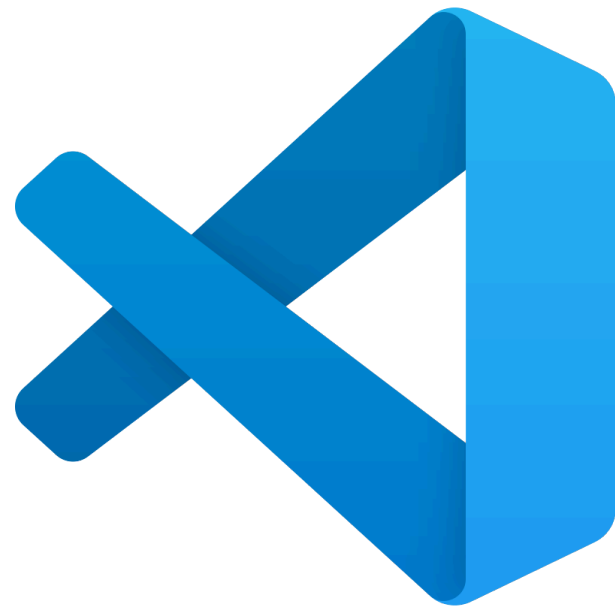
Agent



Allow AI to complete the task

You still ultimately approve the edits, but the AI will edit multiple files, change whatever it needs to.

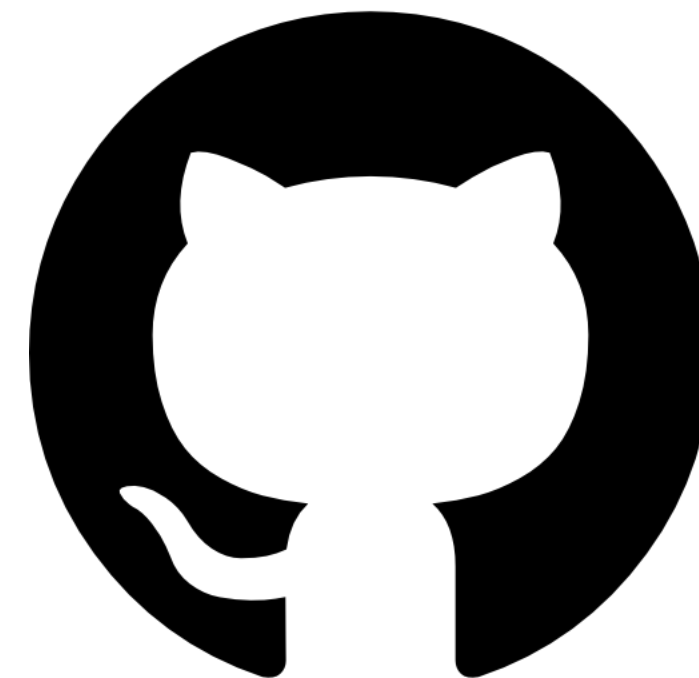
Live Example



VS Code



Co-Pilot



GitHub

<https://github.com/johnny555/ausros.git>